Project Title

myChamber – Web-Based Doctor & Patient Management System

Problem Statement

In Bangladesh, many doctor chambers and small clinics still operate using manual appointment systems. Patients often face difficulties such as:

- Long waiting times at chambers
- Uncertain appointment availability
- No structured way to search doctors by specialization or location
- Frequent no-show appointments causing time loss for doctors
- Lack of secure prepayment system
- Poor record keeping of patient visits

Additionally, there is no unified, chamber-focused digital platform that supports both independent doctors and patients in an organized and secure way.

Therefore, a centralized web-based medical management system is required to improve appointment handling, reduce no-shows, increase transparency, and create a sustainable digital healthcare service platform in Bangladesh.

Objectives of the System

- To digitize doctor chamber appointment management
- To provide an easy doctor search system for patients
- To reduce appointment cancellation and no-show problems
- To ensure secure prepayment before booking
- To create a scalable healthcare business platform

Proposed System Features

User Management

Multi-Role System

- Separate registration and login for:
    - Patients
    - Doctors
    - Admin
- Role-based dashboard access

Profile Management

- Patients can update personal details
- Doctors can manage professional information

Doctor Features

Professional Profile Setup

- Name, qualification, experience
- Specialization category
- Consultation fee
- Chamber location and visiting schedule
- BMDC registration number for verification

Schedule Management

- Add multiple chambers
- Set available days and time slots
- Block unavailable dates

Appointment Handling

- View booked appointments

- Confirm or cancel bookings
- Track booking statistics

Basic Patient History

- View past visits
- Add short consultation notes

Patient Features

Doctor Search & Filter

- Search by specialization
- Search by name
- Filter by location
- Filter by fee range

Appointment Booking

- Select date and time
- Real-time slot availability
- Online prepayment required

Secure Payment System

- Integration with local payment methods (bKash, Nagad, Card)
- Digital receipt generation

Appointment Management

- View upcoming and past appointments
- Cancel appointment (based on refund policy)

Rating & Review

- Patients can rate doctors after completed visits

Admin Features

Doctor Verification

- Approve or reject doctor accounts
- Monitor professional credentials

Platform Monitoring

- Manage users
- Track appointments
- Suspend fraudulent accounts

Revenue Monitoring

- Track total bookings
- Monitor commission earnings
- View system analytics

Business-Oriented Features

- Prepayment system to reduce no-shows
- Commission-based earning model
- Premium doctor listing option
- Refund policy management

Future Improvement Scope

Phase 2 Development

- Telemedicine (video consultation)
- Digital prescription PDF generation
- SMS/Email reminders
- Advanced doctor analytics dashboard

Phase 3 Development

- Mobile application (Android/iOS)
- Electronic Health Record (EHR) system
- Lab test booking integration
- Pharmacy integration
- AI-based smart doctor recommendation